Joint Detection and Active Cancellation of Snoring Signals in Real-Time

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***Abstract*—Snoring noise can be extremely annoying and troublesome, thus influencing people's social lives. Active noise control (ANC) systems can be implemented to reduce unwanted snoring noise, improving people's sleep quality, especially because ANC techniques are more suitable than passive solutions for low-frequency signals, like the snoring signal. However, the intermittent nature of snoring noise may affect the ANC performance. In this context, snoring activity detection (SAD) techniques can be applied to recognize the presence of snoring and accordingly activate the ANC system. The present work proposes a low-latency system based on deep learning for the detection of the snoring activity, combined with an active snoring cancellation (ASC) approach, which uses a delayless subband structure, for real-time applications. Experimental results have been obtained by evaluating the performance of the SAD algorithm, comparing two neural network-based solutions. Finally, the ASC system has been tested with and without the preliminary SAD stage, proving the advantage of using the SAD and ASC algorithms in a joint fashion.**

***Index Terms*—snoring activity detection, active snoring cancellation, deep learning, convolutional neural networks, adaptive subband algorithm**

I. Introduction

Snoring is a significant disorder that can impact people's social lives. The snoring activity produces undesired noise that can have a sound level of 90 dB, causing several consequences such as a decrease in productivity, a reduction of attention, and unsafe driving [1], [2]. The snoring signal contains attributes similar to the vocal signal [3], [4]. The spectrum of both of them is characterized by a fundamental frequency followed by high-order harmonics [4]. The snoring period is composed of two parts: inspiration and expiration. The inspiration is focused on the frequency range between 100 Hz and 200 Hz, and the expiration is contained from 200 Hz to 300 Hz. Therefore, the fundamental frequency to be eliminated is between 100 Hz and 300 Hz.

The noise caused by snoring can be attenuated by several approaches proposed by the literature. The easiest way is to adopt passive solutions, based on the use of physical objects, like earplugs or specific pillows [5]. However, passive techniques show bad performance at low frequencies, and people may find the use of devices very discomforting. The drawbacks of the passive solutions can be overcome by active noise control (ANC) algorithms, which exhibit better performance for low-frequency noise. ANC systems involve a canceling loudspeaker that emits an "antinoise" signal which has an inverted phase with respect to the original noise. In this way, the unwanted signal can be deleted by destructive interference with the added noise in the listening area, monitored by one or more error microphones. ANC approaches can be considered as part of the Internet of Sounds (IoS) field, since they require acquiring and processing of sound signals [6]. Figure 1 shows an example of a single-channel feedforward ANC system. The acoustic path between the noise source, captured by the reference microphone, and the error microphone is called the "primary path", indicated as $p(n)$, whereas the "secondary path", $s(n)$, describes the sound propagation from the canceling loudspeaker to the error microphone. The canceling zone can be enlarged by the use of an array of multiple error microphones, making the system suitable for wireless acoustic sensor networks (WASNs) [7]. Adaptive algorithms should be employed allowing the ANC system to follow the time variations of the noise. The most used technique is based on the filtered-X least mean square (FxLMS) algorithm [8], in which the secondary path estimation is applied to determine the output signal at the error microphone. ANC systems based on FxLMS are developed and applied to snoring cancellation in [1], [9], [10]. A real-time implementation of FxLMS has been recently implemented also in [11] for snoring reduction. The technique has been improved by the integration of the secondary path estimation and a subband structure in [12], [13].

The adaptation procedure can be affected by the fact that snoring is a non-stationary and discontinuous signal. Thus, the performance of the ANC may be reduced by the periods of signal absence. In this context, snoring activity detection is important to identify the presence of a snoring signal and activate the ANC only when it is necessary.

In recent years, deep learning approaches have been widely used to tackle sound event detection tasks, like snoring events. Studies such as [14]–[16] exploited 2D convolutional neural networks (2D-CNNs) leveraging mel-frequency cepstral coefficients (MFCC) computed from fixed-length audio segments, obtaining high values of accuracy. MFCC are extracted by using an analysis window whose size introduces a trade-off between real-time decision-making and capturing longer-term temporal dependencies in snoring patterns. Indeed, longer analysis windows allow for capturing longer-term temporal

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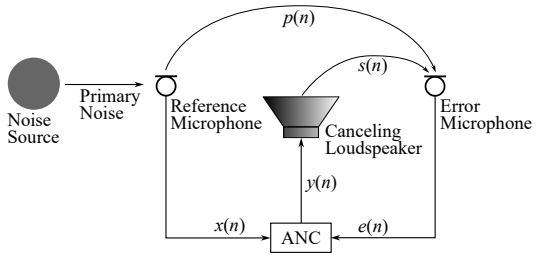


Fig. 1. Single-channel feedforward ANC system.

dependencies, whereas, on the other hand, shorter analysis windows are capable of capturing local temporal dependencies and allow for faster decision responses, making them more suited when real-time constraints have to be met (i.e., real-time audio processing). Recent studies explored the use of sequential models, such as 1D convolutional neural networks (1D-CNNs) [17]–[19] and RNNs [20], [21]. Since they are designed to handle sequential data, they are able to deal with a large temporal context; in this way, they are potentially able to capture longer-term dependencies using shorter analysis windows, leading to faster decision responses, with respect to 2D-CNNs.

Other works like [22] use a feed-forward neural network (FFNN), whereas there are studies exploiting Convolutional recurrent neural networks (CRNNs) that combine the strengths of CNNs and RNNs; in particular, [23]–[25] incorporate a CNN with a long short-term memory (LSTM), whereas [26], [27] use gated recurrent units (GRU) in the recurrent part. It is worth noting that techniques proposed in the literature are not easily comparable, since data on which they are trained and evaluated are often proprietary and have quite different characteristics (in terms of quantity, types, and levels of noises, number of different subjects, etc.). Moreover, the metrics used to assess the performance also vary.

Desirable characteristics for a snoring detection system are good reliability to perform the detection task, the ability to generalize in the presence of various types and levels of noise, and online processing, for systems that should meet real-time requirements, like the one proposed in this paper, in which a successive active snoring cancellation stage takes place.

In this study, we present a joint system for detecting and actively canceling snoring signals. We build upon our previous work [28], by analyzing two detection neural architectures, together with a delayless subband approach for active snoring cancellation. The results show significant improvements for snore cancellation when the snore detection stage is active compared to the situation where it is absent, and also with respect to our previous work.

In this paper, we expand on our prior study [28], reviewing and extending it by presenting significant advancements and analysis. Novel contributions with respect to our previous paper are summarized as follows:

- Introduction of a sequence-to-sequence temporal convolutional neural network (Seq2Seq-TCN), analyzing its performance for the active snoring detection task, also with a comparison with a convolutional recurrent neural network (CRNN), already presented in our previous study [28].
- Data augmentation of the A3Snore (the data corpus already presented in [27], [28]), through the usage of an additional real-world snoring dataset, obtained from the corpus used in [22].
- Snoring detection algorithms now work in a real-time framework; moreover, latency-dependent performance analysis of both CRNN and Seq2Seq-TCN is carried out too.
- Generalization capabilities of both CRNN and Seq2Seq-TCN, and the performance of the overall system have been analyzed on a synthetically generated audio file generated exploiting snoring signals of the Munich Passau Snore Sound Corpus (MPSSC) [29] dataset convolved with a real impulse response measured inside a semi-anechoic chamber, also considering additional noise at various signal-to-noise ratios (SNRs).

The paper is organized as follows. Section II describes the proposed system, focusing on snoring activity detection (SAD) and activity snore cancellation (ASC) algorithms. Section III explains the experimental setup, including datasets used for the evaluation, details related to the system implementation, and evaluation metrics. Section IV shows and discusses the obtained results. Finally, Section V concludes the paper.

II. Proposed system

The scheme of the entire proposed system is shown in Figure 2. The snoring signal $x(n)$ is filtered by the primary path $p(n)$ simulating the sound propagation between the sound source and the error microphone, and then white noise is added to simulate the extra noise that can be generated by the environment, obtaining the signal $d(n)$, which is used as input to the snoring activity detection system. The SAD detects the presence of a snoring signal, producing binary predictions (0 for signal absence and 1 for signal presence). The predictions are sent to the ASC module, which activates or deactivates itself according to the signal presence or absence, respectively. Detailed explanations of the involved algorithms are reported in the following subsections.

A. Snoring activity detection

For the snoring activity detection stage, we considered two different neural architectures, a Convolutional Recurrent Neural Network (CRNN) and a Sequence-to-Sequence Temporal Convolutional Neural network (Seq2Seq-TCN). The former was at first proposed in [27], and also implemented in our previous work [28], whereas the usage of the latter comes from recent works of Luo et al. [30], [31], which tackled the snoring detection task by using this kind of neural architecture. Snoring detection is inherently a time-series classification task, and both aforementioned architectures are well-suited for capturing temporal dependencies from a signal that has a sequential nature like snoring.

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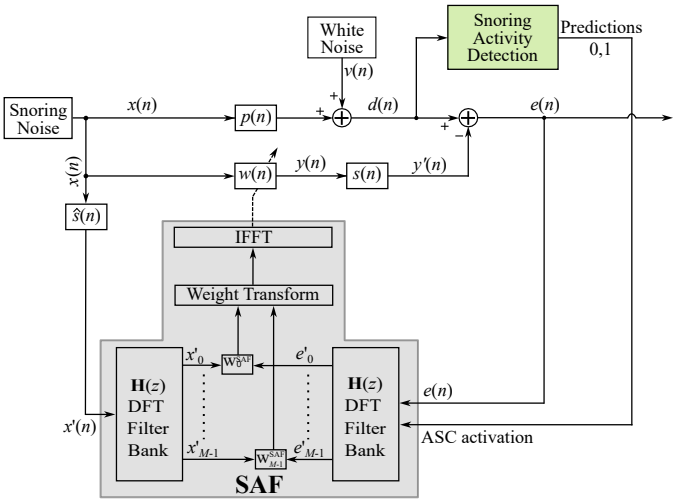


Fig. 2. Scheme of the proposed system.

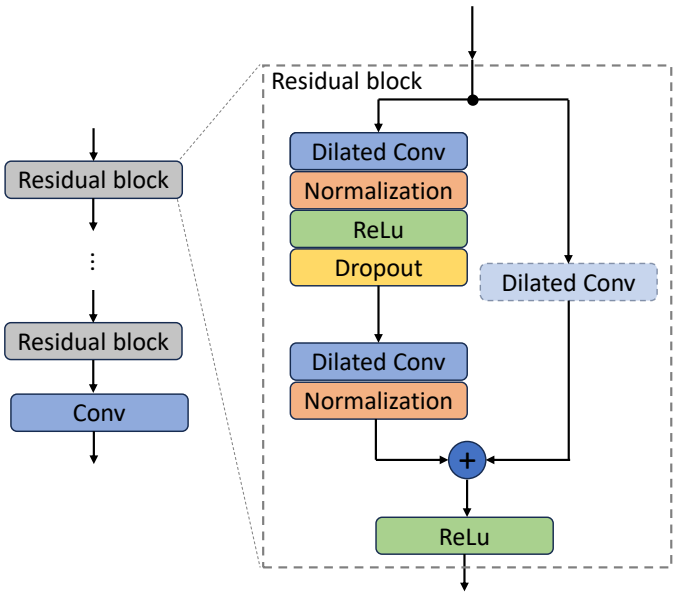


Fig. 4. Architecture of the Seq2Seq-TCN under study.

The CRNN has the same structure as the one proposed in Figure 3. It has an initial convolutional part aimed at learning patterns in features, followed by recurrent layers aimed at catching temporal dependencies; it uses time-distributed layers in order to process data maintaining, at the same time, temporal information.

On the other hand, the Seq2Seq-TCN has the architecture depicted in Figure 4. It is composed of a series of residual blocks and a convolutional layer as the final layer. Each residual block has the structure presented in Figure 4. As it can be seen, it consists of two consecutive groups of layers; the first group includes a convolutional layer, a normalization layer, a Rectified Linear Unit (ReLU) [32] activation function, and a dropout, whereas the second group implements just a convolutional layer and a normalization layer. The introduction of the ReLU activation function is useful in the optic of handling non-linearities, the normalization layer prevents the explosion of the gradients, whereas dropout acts as a regularization technique to reduce the risk of overfitting. Moreover, as it can be seen in Figure 4, there is also a residual connection, therefore, the output of the residual block corresponds to the output of the final ReLU activation function

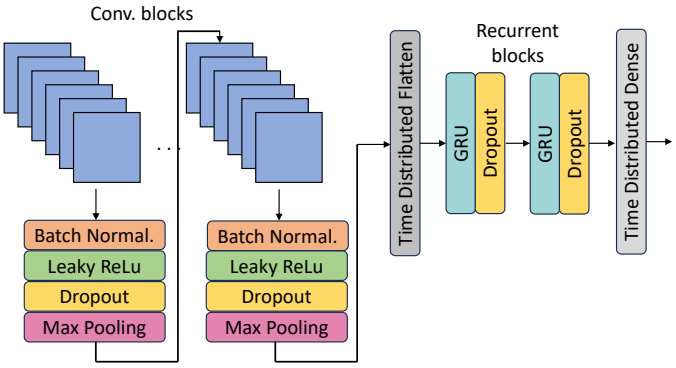


Fig. 3. Architecture of the CRNN under study.

that takes as input the result of the element-wise addition of the residual connection and the pipeline of the two aforementioned groups of layers. For almost all blocks residual connections are just skip connections, except for the first block; indeed, it also implements a convolutional layer, depicted as slightly transparent and with dashed borders in Figure 4.

B. Active snoring cancellation

The active snoring cancellation (ASC) system is obtained by a feed-forward filtered-x structure with subband adaptive filtering, as proposed in [13]. Figure 2 shows the scheme of the algorithm. The input signal $x(n)$ represents the snoring signal captured by the reference microphone, located close to the noise source. The signal $e(n)$ is the one recorded by the error microphone, placed in the zone where the snoring noise must be deleted and it is computed as

$$e(n) = d(n) - y'(n) = d(n) - s(n) * y(n), \tag{1}$$

where $d(n)$ is the desired signal, $s(n)$ is the secondary path, and $y(n)$ is the "antinoise" signal generated by the canceling loudspeaker, defined as

$$y(n) = w(n) * x(n), \tag{2}$$

where $w(n)$ is the adaptive filter that estimates the primary path $p(n)$. The coefficients of the adaptive filter are obtained through a delayless subband adaptive filtering (SAF) structure, first proposed in [33] and then applied to ANC in [12], [13]. The signal $x'(n)$, which is the input $x(n)$ filtered by the estimated secondary path $\hat{s}(n)$, and the error $e(n)$ are decomposed in subbands by the analysis filterbank $\mathbf{H}(z)$, described as

$$\mathbf{H}(z) = [H_0(z), H_1(z), ..., H_{M-1}(z)]^T, \tag{3}$$

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where $H_m(z)$ is the transfer function of the k-th analysis bandpass filter of length N_p, with $m = 0, ..., M-1$, and M the number of subbands. After the decomposition, the signals $x'_m(n)$ and $e'_m(n)$ for each m-th subband are obtained. The weights of the m-th subband filter $\mathrm{w}_m^{\mathrm{SAF}}(n)$ at the time step $n+1$ are calculated following the normalized least mean square (NLMS) algorithm as

$$\mathrm{w}_m^{\mathrm{SAF}}(n+1) = \mathrm{w}_m^{\mathrm{SAF}}(n) + \mu_w \frac{x'^*_m(n) e'_m(n)}{\alpha + ||x'_m(n)||^2}, \qquad (4)$$

where $x'^*_m(n)$ is the complex conjugate of $x'_m(n)$, μ_w is the step size, and α is a small coefficient that avoids division by zero. Finally, the fullband filter $w(n)$ is attained by implementing the following steps:

- the subband weights are transformed by the calculation of M fast Fourier transforms (FFTs) of length N/D, where $D = M/2$ is the decimation factor, and N is the length of the fullband filter $w(n)$;
- the complex samples of FFT are stacked to form the first half of the array of the fullband filter;
- to complete the array, the central point is set to zero, and the second half of the array is obtained by complex conjugating and reversing the first half of the array;
- the inverse FFT (IFFT) of length N is performed to obtain the fullband filter $w(n)$.

The update algorithm and the subband filtering operate when the snoring detection algorithm reveals the presence of snoring.

III. Experimental setup

A. Datasets

1) Snoring activity detection: The training phase of both snoring detection neural models has been conducted by exploiting two different datasets containing real-world snoring signals.

The first, A3Snore, is the one introduced in [27] and also considered in our previous work [28]; it contains whole night recordings of two subjects divided into audio files 10 minutes long. Files without snores have been discarded, therefore it includes a total audio amount of 6 h 50 min, with 51 min of snores. Audio files have been recorded in wav format with a sampling rate of 44.1 kHz, and they were manually annotated. Details regarding the recording setup and scenario can be found in [27].

The other dataset has been obtained from the corpus used in [22]. It contains 5000 variable-length audio samples, equally distributed among snoring and non-snoring classes, coming from recordings of 50 different subjects, for a total audio amount of 1 h 59 min, with 1 h 10 min of snores.

Whereas A3Snore already includes audio samples containing both snoring and non-snoring, for what concerns the other dataset, we concatenated (with cross-fading to avoid clipping) its snores and non-snores audio samples (in particular, one snore followed by a variable number of non-snores), in order to obtain 50 longer audio files, one for each subject. Then, we performed feature extraction on the audio files of each dataset, by computing 40 MFCC considering a window size of 30 ms and a hop size of 10 ms. In this way, we obtained a set of feature arrays, one for each audio file; then, we split each of them into shorter equal-length sequences. Since the two datasets are characterized by different statistics they need to be correctly managed before being used. Therefore, we first separately standardized sequences of features associated with both data corpora; therefore, only after standardization, we concatenated them. In this way, the final dataset will have a distribution with zero mean and unitary standard deviation. It is worth noting that inverting these steps, i.e., standardization and concatenation, leads to low training performance, as experienced in our simulations. Finally, the overall set of MFCC sequences obtained by transforming the audio frames is then split into training, validation, and testing subsets with a ratio of, respectively, 80/10/10.

2) Active Snoring Cancellation: The input signal used for testing the ASC algorithm has been obtained by the concatenation of the snoring excerpts of the MPSSC dataset (a data corpus of 828 audio files extracted from 219 subjects) [29] with silence frames of different casual duration, obtaining the input signal $x(n)$ depicted in Figure 2. Then, the signal $x(n)$ is convolved with the primary path impulse response $p(n)$, and white noise with different gains has been added, obtaining the signal $d(n)$ for different values of SNR. The signal $d(n)$ has been finally used as input to the snoring activity detection system and as the desired signal by the ASC algorithm, as reported in Figure 2.

B. System implementation details

1) Snoring activity detection: Regarding snoring activity detection, we considered the two neural architectures already presented in Section II-A. For CRNN, we considered three consecutive convolutional blocks, each consisting of a convolutional layer, a batch normalization layer, a dropout layer, and a max pooling layer. In each block, convolutional layers have 64 filters with size (3, 3); a Leaky ReLU activation function is applied to their output. Filters in max-pooling layers decrease their size at each block, from (5,1) to (4,1) to (2,1). The output of the convolutional part is flattened in order to be passed as input to the recurrent part of the CRNN. This considers two blocks, each one with a GRU layer with 64 units, and a dropout layer. A rate of 0.3 is considered for all dropout layers. Its number of parameters is about 120k. The training has been carried out by using the early-stopping strategy and a batch size equal to 8.

On the other hand, for the Seq2Seq-TCN, we considered 1D convolutional layers with eight filters, a kernel of size 3, and batch normalization layers. The number of residual blocks J has been computed as follows:

$$J = \left\lceil \log_b \left(\frac{(L-1)\cdot(b-1)}{(K-1)\cdot 2} + 1 \right) \right\rceil \qquad (5)$$

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where b is the dilation base equal to 2, L is the input sequence length equal to 1500, and K the kernel size, obtaining J equal to 9.

This equation ensures that the number of residual blocks is the minimum that guarantees the "full history coverage" condition [34]. The dilation rate considered in each successive residual block increases following the power of 2. The number of network parameters results in about 5k, therefore more than 20 times fewer than the CRNN-based method. The training has been carried out by using the early-stopping strategy and a batch size equal to 4. Both neural networks have been implemented with the TensorFlow deep learning framework[1], and experiments have been carried out on an NVIDIA DGX Station A100, by exploiting its GPUs.

In particular, the training phase has been performed similarly for both networks by passing them disjointed sequences of MFCC.

On the other hand, inference has been carried out on the test set, extracted from the dataset used for training (as described in Section III-B), by using overlapped sequences as input in order to deal with limited values of latency. In particular, we dealt with latency values down to 10 ms.

Moreover, we also performed inference on new synthetically generated signals, containing non-snoring intervals separated by an interval containing the result of the temporal convolution of a real-world snore of the MPSSC dataset and the impulse response measured inside a semi-anechoic chamber, and also additional noise at various SNRs.

Downstream of the neural network that performs the frame-level classification of snore and non-snore events, we considered a post-processing stage in order to refine output predictions with the aim of reducing the number of false negatives and facilitate the convergence of the active snoring cancellation algorithm. This stage has been implemented as a Hangover algorithm, and we already presented it in [28]. It is based on a majority voting scheme, therefore it exploits an input buffer with odd size, equal to $X = 3$ frames in our case, that acts as a FIFO (First-In First-Out) register, and each 10 ms gets a new sample. Each time the buffer contains more ones than zeros, the Hangover algorithm, in order to decrease the number of false negatives, sets the next $k = 30$ consecutive predictions to one; in this way, it allows to obtain an output sequence of predictions more suited, i.e., smoother, for the ASC stage.

2) Active snoring cancellation: The ASC algorithm has been evaluated by comparing its performance with and without the presence of the SAD system. Without SAD, the ASC continuously updates the filter coefficients of $w(n)$, according to Equation (4). Differently, when SAD is operating, the filter's update of the subband structure is performed only when snoring noise is detected, i.e., when the prediction is 1. In particular, when the ASC system receives the first true prediction, it starts the adaptation and continues for 5 seconds. Since the snoring excerpts are shorter than 5 seconds, it means that, in the first period, the adaptation is carried out also in parts of the signal where only white noise is present. This time has been chosen after several experimental tests to improve the performance of the algorithm. After this interval, the update is disabled when the ASC system receives a 0 and is enabled in case of a 1.

The primary path $p(n)$ and the secondary path $s(n)$ are measured in a semi-anechoic chamber considering the setup of [11] and are designed by FIR filters with a length of 256 samples. The noise source is positioned on the left side of the bed, the reference microphone is located near the noise source, the canceling loudspeaker is mounted on the right side of the head of the bed at 90 cm from the primary noise source, and the error microphone is 40 cm to the right of the canceling loudspeaker [11]. The adaptive filter $w(n)$ has a length of $N = 512$ samples. The subband structure has been derived from a prototype FIR filter with a length of 256 samples, and the number of the subbands is set to $M = 64$. The step size used for the adaptation algorithm is $\mu = 0.05$.

C. Evaluation metrics

For the snoring activity detection stage, we considered the Average Precision (AP) - also known as the area under the precision-recall curve (AUC-PR) - as the evaluation metric. It has been selected with respect to other metrics usually considered for binary classification tasks, such as accuracy, due to class imbalance issues. The Average Precision is defined as follows:

$$AP = \sum_{th} (R_{th} - R_{th-1}) \cdot P_{th} \tag{6}$$

where R_{th} and P_{th} are, respectively, the recall and precision computed for the threshold th.

Regarding the ASC algorithm, the estimated response has been compared with the measured one by evaluating the log-spectral distance (LSD) between the frequency response of the measured primary path $P(k)$ and the frequency response of the estimated primary path $W(k)$, as [35]

$$\text{LSD} = \sqrt{\frac{1}{k_2 - k_1 + 1} \sum_{k=k_1}^{k_2} \left[10 \log_{10} \frac{\left|P(k)\right|^2}{\left|W(k)\right|^2} \right]^2}, \tag{7}$$

where k_1 and k_2 delimit the frequency range within which the LSD is estimated, defined as $B = [k_1 \frac{f_s}{F}, k_2 \frac{f_s}{F}] = [100\,\text{Hz}, 20\,\text{kHz}]$, with $F = 4096$ the number of frequency bins for the FFT computation, and $f_s = 44.1$ kHz the sampling frequency.

The obtained results are reported in the following section, comparing the performance without SAD and when SAD system is active, for different SNR values of the input signal $d(n)$.

IV. Results and discussion

The experimental results are obtained by calculating the evaluation metrics discussed in Section III-C, testing the snoring activity detection and the active snoring cancellation algorithms.

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A. Snoring Detection

Figure 5 reports results obtained by performing inference with both networks on the test set, obtained by the split performed on the dataset used for training (as described in Section III-B), considering the model performing the best on the validation set. Results are expressed in terms of Average Precision, computed for various latency values. One frame corresponds to 0.01 s, therefore, latency values considered range from 0.01 s to 15 s. As can be seen, CRNN outperforms

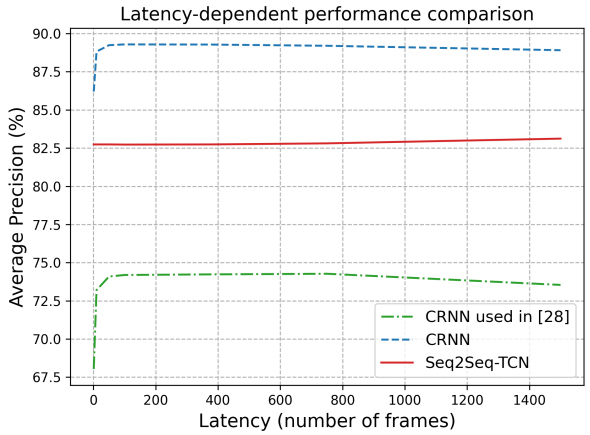


Fig. 5. Average Precision plotted by varying latency for both SAD systems presented in this study, and the CRNN proposed in [28].

the Seq2Seq-TCN for all considered values of latency. The plot shows that a latency relatively small, i.e., 100 ms, is enough to allow the CRNN to reach an Average Precision of 88.82 %, near its maximum (89.29 %). For what concerns the Seq2Seq-TCN, it pretty maintains the same performance by varying the latency. The plot shows that, for both networks, a small latency gives sufficient temporal context to capture relevant information from the input features, and that additional latency did not necessarily improve, or only for a small amount, the performance; therefore, it is preferable to work with the lowest latency. Results have been also reported for the CRNN used in [28], trained on the A3Snore only, which shows lower performances than the other two.

In order to also have additional visual feedback of the performance of the overall active snoring detection system, we depicted in Figure 6 the predictions generated by the Seq2Seq-TCN before (raw) and after (post-processed) the application of the Hangover algorithm, for a 50-s excerpt of the audio signal generated starting from MPSSC audio files, as described in Section III-A2, for the case SNR = 20 dB.

Figure 7 shows the predictions post-processed by the Hangover algorithm along with the input signal with SNR = 20 dB, for a 100-s audio excerpt. In particular, in Figure 7(a), the post-processed predictions obtained performing snoring detection with the CRNN trained with the dataset described in Section III-A are compared with the ones obtained when using the CRNN presented in [28] (trained with A3Snore dataset only). As can be seen, the CRNN trained with the dataset presented in Section III-A allows to better identify snoring events. Figure 7(b) compares the post-processed predictions obtained after performing snoring detection with the two networks, i.e., CRNN and Seq2Seq-TCN, presented in this study, showing that both of them are able to correctly identify snoring events.

B. Active snoring cancellation

The results of the ASC system have been evaluated in terms of LSD, following Equation (7). The experiments for the ASC system consider the predictions obtained with the lowest latency of 0.01 s and post-processed by the Hangover algorithm. Table I shows the values of the LSD for different values of SNR of the input signal, comparing the ASC when SAD is disabled, the ASC with SAD performed by Seq2Seq-TCN, and the ASC with SAD when CRNN is applied. It can be seen that when SAD is activated, the estimation performance of the ASC system improves. The greatest improvement is reached for lower values of SNR up to 25 dB. Differently, when the SNR is higher than 25 dB, the ASC shows good LSD values for all three cases. In particular, the ASC system consistently achieves the lowest LSD values for all evaluated SNRs when snoring activity detection is performed by the Seq2Seq-TCN; this outcome could be attributed to the fact that the predictions obtained by Seq2Seq-TCN are more synchronized with the snoring events of the input signal since temporal alignment is important for the FxLMS-based ASC system to apply precise adaptive filtering to effectively reduce the snoring signal.

An example of the estimated primary path is shown in Figure 8, where the measured primary path is compared with the one estimated by the ASC system when SAD is disabled (i.e., SAD OFF), the one estimated by the ASC stage with SAD activated (i.e., SAD ON) and performed by Seq2Seq-TCN, and the one estimated by ASC stage with SAD performed by CRNN, both in the time domain (cf. Figure 8(a)) and in the frequency domain (cf. Figure 8(b)), considering an input signal with SNR $=$ 20 dB. The main peak of the impulse response is well-detected in all cases, however, the differences are more clear looking at the frequency responses, depicted in Figure 8(b). As proved by the LSD values of Table I, the presence of SAD improves the estimation, and the best performance is achieved by the Seq2Seq-TCN. This result is more evident at the low-frequency part of the spectrum up to 5 kHz, where most of the signal energy is located.

V. Conclusions

In this paper, a joint system for real-time detection and active cancellation for snoring signals has been proposed.

As an evolution of our previous work, the proposed study is now able to work in a real-time framework. It also includes other several advancements, such as the implementation and comparison of two neural-based snore detection systems, i.e., CRNN and Seq2Seq-TCN, trained on the same data, and the data augmentation performed on the A3Snore corpus carried out by exploiting an additional real dataset. Moreover, the performance of both methods has been assessed in terms

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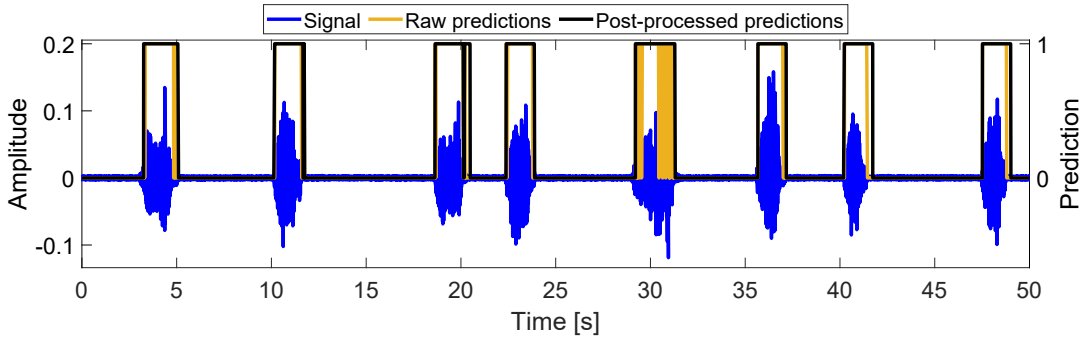
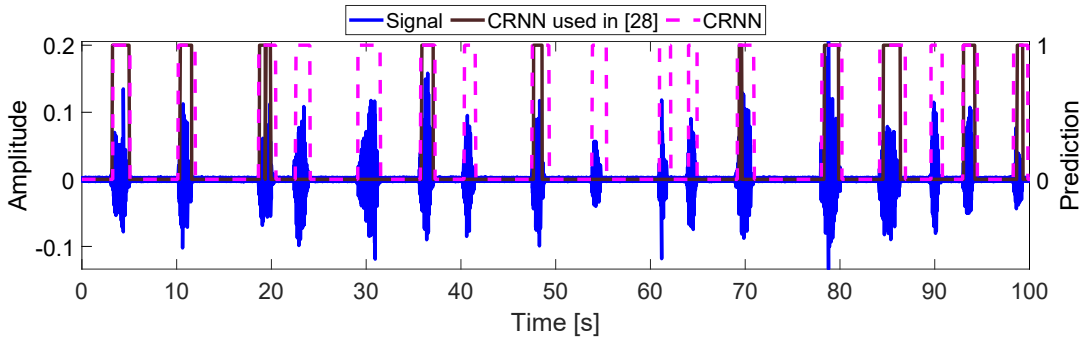
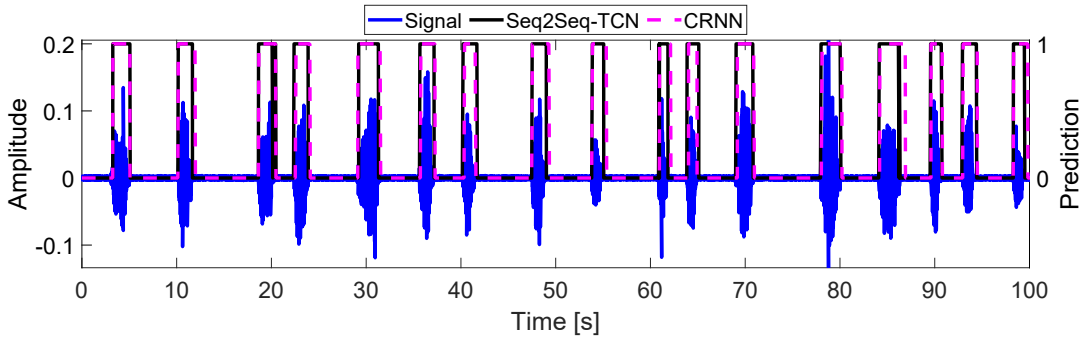


Fig. 6. Input signal with SNR = 20 dB (blue), and both the raw (yellow) and post-processed (black) predictions obtained after having performed snoring detection with the Seq2Seq-TCN.



(a)



(b)

Fig. 7. Input signal with SNR = 20 dB is depicted in blue. (a) shows the comparison between the post-processed predictions obtained after performing snoring detection with the CRNN used in [28] (i.e., trained on A3Snore only) and the CRNN trained with the dataset presented in this paper, whereas (b) compares post-processed predictions obtained after having performed snoring detection with Seq2Seq-TCN and CRNN, both trained with the dataset presented in this study.

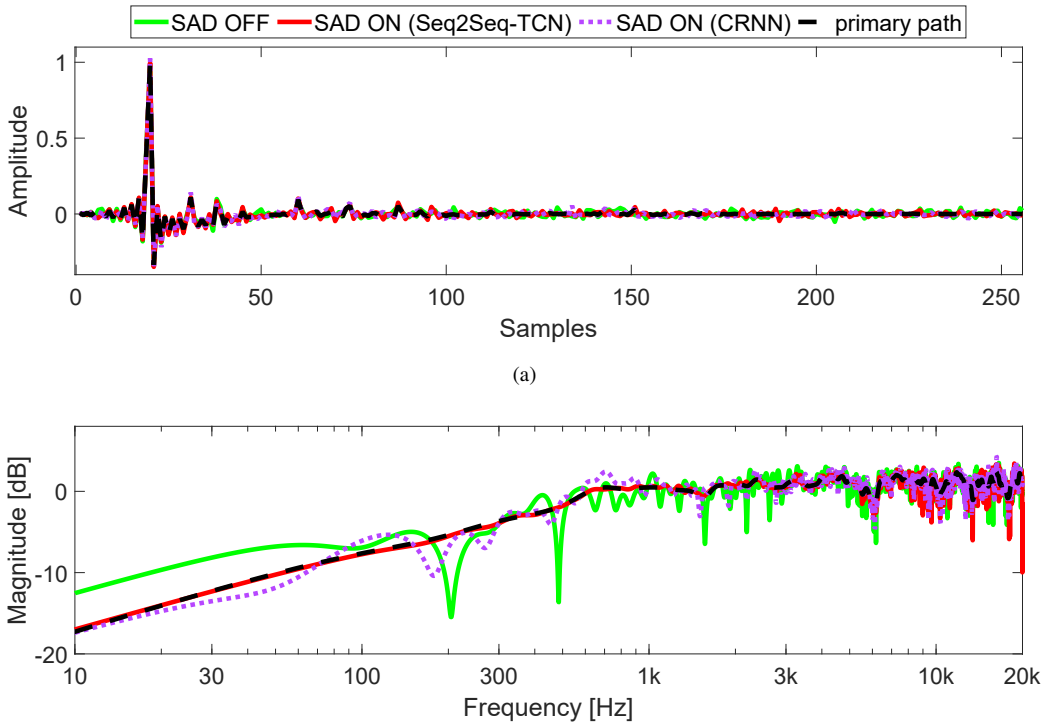
of Average Precision with a latency-dependent analysis performed on the test set, extracted from the dataset used for training (as described in Section III-B).

The ASC is developed by implementing a delayless subband structure to maintain a reduced latency of the total system, suitable for real-time applications.

The overall system has been evaluated by taking into consideration an audio file generated with MPSSC dataset's audio files, a real impulse response measured in a semi-anechoic chamber, and various levels of SNR. The Seq2Seq-TCN-based snoring detection technique has shown lower performance than the CRNN on the test set extracted from the dataset used for training, whereas both networks are able to work well on the synthetically-generated audio file. When performing snoring cancellation on the synthetically-generated file, the ASC system performs better when it is used in combination with Seq2Seq-TCN; this may be due to better temporal synchronization. Thanks to its lower number of parameters (i.e., more than 20 times fewer), Seq2Seq-TCN is also well-suited for edge-computing-oriented applications.

For future works, the proposed system can be adapted for WASN applications. By utilizing microphone arrays, the

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(a)

(b)

Fig. 8. Comparison between the measured primary path, the primary path estimated in the case of SAD OFF, SAD ON with Seq2Seq-TCN and SAD ON with CRNN, (a) in the time domain and (b) in the frequency domain, considering an input signal with SNR = 20 dB.

TABLE I
VALUES OF THE LSD CONSIDERING SAD OFF AND SAD ON, WITH PREDICTIONS GENERATED BY SEQ2SEQ-TCN AND CRNN, CONSIDERING THE LOWEST LATENCY OF 0.01 S, FOR DIFFERENT SNR VALUES OF THE INPUT SIGNAL. THE LSD IS CALCULATED IN THE FREQUENCY RANGE OF [100 HZ–20 KHZ].

LSD [dB]			
SNR [dB]	SAD OFF	SAD ON (Seq2Seq-TCN)	SAD ON (CRNN)
10	1.44	**1.08**	1.11
15	0.86	**0.63**	0.67
20	0.50	**0.33**	0.45
25	0.27	**0.20**	0.23
30	0.15	**0.13**	0.14
35	0.09	**0.07**	0.09
40	0.07	**0.06**	0.07

system can benefit from sound source localization algorithms to enhance SNR, therefore facilitating the snoring detection task. The deployment of strategically distributed multiple microphones allows the canceling zone to be enlarged and this can lead to improved snoring cancellation.

The deployment of our system into a real-world WASN requires careful consideration of several factors, such as latency, which plays a central role in achieving real-time snoring cancellation. The decision-making strategy could follow either a decentralized or a centralized approach [36], [37]. The former employs distributed edge computing across dislocated sensors, enabling local pre-processing tasks (e.g., noise filtering, feature extraction) at each node. This reduces bandwidth requirements and the computational burden on the central unit; in this way, it decreases the risk of network congestion and allows for lighter communication protocols, leading to reduced latency. However, it demands rigorous sensor synchronization [38], potentially increasing overhead. Conversely, centralized approaches, which are based on a central unit, simplify sensor synchronization requirements, potentially decreasing overhead. However, as all sensors transmit data to the central unit, the risk of network congestion is higher, and it can be mitigated by increasing overhead, resulting in greater latency. Energy-related constraints should also be accounted for since WASN sensor nodes are usually battery-powered. The selection of the appropriate decision-making strategy and the network protocol depends on the specific application requirements and resource constraints.

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